Failure Analysis Report

FAIL-0734 — Resource Exhaustion

1. Failure Summary

Failure ID	FAIL-0734
Failure Type	Resource Exhaustion
Severity	Major
Component	Media Codec
Application	APP-005 — WearableHealth
Build Version	5.1.1
Timestamp	2024-11-14 06:38:21
Environment	ENV-002 — Wearable Lab
OS Version	Wear OS 4
Network	Bluetooth 5.3
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-2098
Title	Confirm that firmware over-the-air update triggers the expected notification whe...
Execution ID	EXEC-03823
Test Run	RUN-007 — Regression Cycle 7 — Build 5.3.1
Requirement	REQ-0457 — Data Synchronization — Remote Diagnostics

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete health metric upload within the defined SLA.
Observed Behaviour	The system exhibited resource exhaustion when the target device is running a legacy firmware version...

4. Error Details

Error Message:

```
OOMError: heap exhausted at 1380MB/2048MB during media_decode.
```

Stack Trace / Log Excerpt:

```
com.samsung.smarthome.VoicePipeline.execute(line 594)
```

Log Reference	LOG-64423

5. Executed Steps at Time of Failure

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1	Re-connect the wearable and confirm auto-recovery.	The audit log records the event with correct metadata.
2	Disconnect the wearable and wait 22 seconds.	Server returns HTTP 200 with a valid JSON payload.
3	Enter valid credentials: username='testuser_023', password='[REDA...	No crash or ANR is observed.
4	Disconnect the headset and wait 22 seconds.	The device reconnects automatically within 10 seconds.
5	Enter valid credentials: username='testuser_026', password='[REDA...	A confirmation notification appears on the paired device.

6. Root Cause Analysis

Missing null-check on the device registry response before dereferencing the session token.

8. Historical Similar Failures

HIST-0427	FAIL-0428	Resource Exhaustion	4.0.1	2022-08-11	9	Resolved
HIST-0636	FAIL-0637	Resource Exhaustion	6.3.2	2023-07-20	8	Resolved
HIST-0650	FAIL-0651	Resource Exhaustion	3.2.5	2021-08-29	4	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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